

Meghan Tennant

In this module, we learned all about programming and how to program a microcontroller to complete tasks. We started this module off learning about the general programming approach and how to plan out the implementation, using flow charts and system diagrams. We continued to use these skills throughout the remaining classes of this module. We also created our Github accounts, which we utilized to organize and share our programming activities. We also worked on our portfolios, which will be very helpful in the future when we're looking for internships and jobs. In the next lecture, we explored Arduino and the Arduino Nano ESP32, looking at the pinout or the output map for all pins. We also utilized MicroPython to code these, learning important functions such as print, pin, value, and sleep.

During the next lecture, we used solderless breadboards to build various circuits that we connected to Power Supply. We experimented with voltage, turning the pot resistance all the way down to purposely blow our resistor. We constructed a few circuits, one with a switch and another with a light sensor. We programmed the microcontroller with the help of Gemini, giving the AI a prompt to generate code for MicroPython based on what we wanted to happen. The last day, we built a circuit with a moving servo and then focused on one with a rain drop sensor. We recorded both manual and automated data for this circuit, based on how wet the sensor was, and used google sheets to analyze the data. We found the averages and standard deviations of both the manual and automated data, and then plotted that information on a scatter graph.

I learned a lot of Github terminology that I never knew before, which will be helpful as I continue my journey in this program throughout college. We also explored Arduino, a program that we used to control the microcontroller through coding MicroPython. My previous knowledge of Python definitely helped me understand MicroPython better, but it was still very new to me. I also learned about solderless breadboards, including how to build a circuit on one and how to power it. I used this the

following day when we constructed circuits with rain drop sensors, recording data across multiple trials in Google Sheets. I learned how to calculate standard deviation and average using built-in functions, which I will definitely be using in the future. Overall, I feel that my group worked well together, even better than the previous module.